VR-CLIP: Visual Refinement of CLIP for Zero-Shot Semantic Segmentation

Anonymous CVPR submission

Abstract

Zero-shot semantic segmentation has gained significant attention with the emergence of vision-language models such as CLIP. Most existing methods primarily focus on improving generalization or designing more efficient decoders. However, they remain constrained by CLIP's image-level visual features, which limit pixel-level discriminability and make dense predictions heavily reliant on the decoder. To address these issues, we are the first to explore optimizing visual features for zero-shot semantic segmentation task. Building on this idea, we propose ***VR-CLIP*** *(Visual Refinement of CLIP), a one-stage framework that refines CLIP's visual representations within the encoder from both semantic and spatial perspectives. Specifically, the Neighborhood Spatial Awareness (****NSA****) module enhances visual continuity through local spatial modeling, while the Semantic-Guided Visual Refinement (****SGR****) integrates class-level semantic cues to strengthen feature discriminability and preserve CLIP's inherent visual-linguistic alignment. Moreover, a parameter-sharing design within SGR further promotes generalization to unseen categories. Extensive experiments on PASCAL VOC 2012, COCO-Stuff 164K, and PASCAL-Context demonstrate that VR-CLIP consistently outperforms state-of-the-art methods under both inductive and transductive settings.*

1. Introduction

Semantic segmentation is a fundamental task in computer vision that assigns a semantic label to each pixel in an image [5, 7, 8, 14, 16, 38]. Although classical methods have achieved remarkable success, they heavily rely on large-scale datasets with pixel-level annotations [4, 12, 31, 43, 48], the collection of which is both costly and labor-intensive. To alleviate this dependency, low-supervised approaches have been explored. Among them, zero-shot semantic segmentation (ZS3) [1, 13, 36] is particularly noteworthy, as it enables segmentation of unseen categories solely based on semantic descriptions, effectively overcoming the limitations of traditional closed-set models.

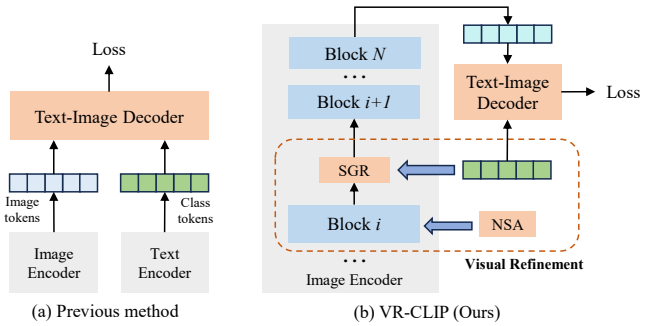

Figure 1. (a) Previous one-stage methods. (b) Our VR-CLIP introduces two designs into the CLIP image encoder to refine visual features while preserving visual-language relevance.

In recent years, the emergence of the pre-trained vision-language model CLIP [29] has introduced a new paradigm for ZS3. Its strong cross-modal alignment capability has been effectively extended to dense prediction tasks, driving remarkable progress in ZS3.

Early works such as ZSSEG [39] and ZegFormer [9] adopted a **two-stage** pipeline that generates mask proposals and then classifies them using CLIP. Although effective, this design is computationally expensive and fails to fully exploit CLIP's potential for ZS3. ZegCLIP [47] later pioneered a **one-stage** architecture (Fig. 1(a)) that performs pixel-level segmentation by directly matching text embeddings with features from CLIP's final visual layer. [18, 41, 45] enhance generalization through distinct decoder designs, Cascade-CLIP [23] matches multiple fixed intermediate-layer features through cascaded decoders to capture richer object details, CLIP-RC [44] leverages regional clues to connect image and pixel-level semantics, and TagCLIP [22] improves text discriminability via trusty tokens. Most existing methods primarily focus on improving generalization or designing more efficient decoders. However, they remain constrained by the image-level nature of CLIP's visual features, relying heavily on the decoder to achieve dense predictions. Specifically, the original CLIP visual features are optimized for global image–text alignment, causing the model to overemphasize the most class-

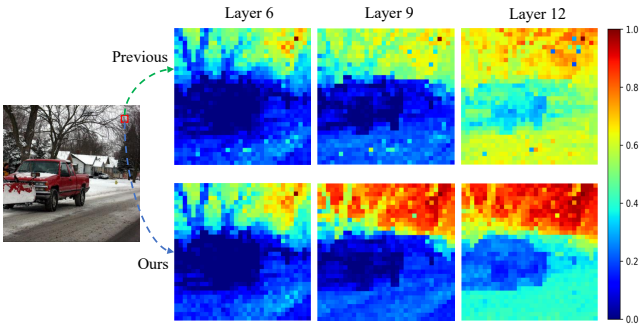

Figure 2. For unseen class tree, we visualize the cosine similarity maps of visual features from different layers of the visual encoder. The previous method tends to focus only on the most similar local regions, whereas our approach exhibits stronger semantic continuity within the same class and better discrimination across different classes, particularly in the final-layer (layer 12) visual representations.

discriminative regions, which in turn leads to more discrete feature associations within the same object (Fig. 2). Consequently, the decoder struggles to establish precise pixel–text correspondence, leading to incomplete or fragmented segmentation results, especially in complex scenes. These limitations motivate us to explore optimizing the visual representations for ZS3 task. This represents an unexplored approach distinct from prior works.

Several works [6, 19, 24, 32] have explored adapting CLIP's visual features for dense prediction tasks, such as fine-tuning or training-free adaptation methods. However, these approaches are not directly applicable to ZS3 due to the inherent limitations. First, fine-tuning CLIP on seen classes compromises its generalization ability, leading to significant performance degradation on unseen classes. More importantly, most of these approaches are asymmetric, where the visual encoder shifts its alignment objective from image-to-text to pixel-to-text, while the text encoder remains aligned at the image level. This asymmetry inevitably disrupts CLIP's inherent cross-modal consistency, which is critical for accurate pixel–text matching in ZS3 decoders. These issues highlight a fundamental challenge: enhancing pixel-level discriminability while preserving both generalization to unseen classes and CLIP's inherent visual–linguistic alignment.

Building upon these observations, we propose VR-CLIP, a one-stage framework designed to enhance CLIP's visual encoder for ZS3. Our method refines visual features within the encoding stage, preserving CLIP's inherent visual–linguistic alignment while enriching pixel-level representations, as shown in the Fig. 1(b). Specifically, we introduce a lightweight Neighborhood Spatial Awareness (NSA) module that leverages the local perception capability of convolution to alleviate the dominance of global representations caused by attention mechanisms. Simultaneously, leveraging the visual–language alignment of the pre-trained CLIP, we propose Semantic-Guided Visual Refinement (SGR) that integrates semantic features into visual representations through semantic similarity maps, enabling joint visual and linguistic perception. Furthermore, parameter sharing across partially common training layers enhances the model's generalization to unseen categories. Through the joint modeling of spatial and semantic cues, VR-CLIP effectively bridges the gap between CLIP's image-level understanding and pixel-level prediction, achieving strong performance on both seen and unseen classes. In summary, our contributions are listed as follows:

- We are the first to explore refining visual representations for ZS3, and propose VR-CLIP to overcome the limitations of CLIP's image-level features and enhance pixel-level discriminability.
- We propose two complementary modules, Neighborhood Spatial Awareness (NSA) and Semantic-Guided Visual Refinement (SGR), which inject spatial and semantic cues into visual features, while preserving both generalization to unseen classes and CLIP's inherent visual–linguistic alignment.
- We validate VR-CLIP on three benchmark datasets under both inductive and transductive settings, demonstrating consistent improvements over state-of-the-art methods on both seen and unseen classes.

2. Related Works

2.1. Pre-trained Vision Language Models

Pre-trained Vision-Language Models (VLM) learn aligned representations between visual and textual modalities through large-scale contrastive training. Among them, CLIP [29] has become the most influential foundation model, jointly optimizing an image encoder and a text encoder on massive image-text pairs to achieve strong zero-shot transfer across diverse vision tasks. Its success has inspired numerous extensions [17, 20, 21, 34, 42], demonstrating that large-scale multimodal pre-training can serve as a universal representation learner. These models exhibit remarkable generalization in downstream tasks, providing a robust prior for zero-shot visual understanding.

2.2. Zero-Shot Semantic Segmentation

ZS3 aims to segment unseen classes using only annotations of seen classes, addressing the scalability limitations of closed-set segmentation models. Early approaches focused on aligning visual and semantic spaces: SPNet [37] proposed semantic projection to transfer knowledge from seen to unseen classes, while ZS3Net [1] and CaGNet [13] leveraged word embeddings and context-aware feature generation to improve generalization. STRICT [27] introduced self-training with pseudo-labels for transductive ZS3, miti-

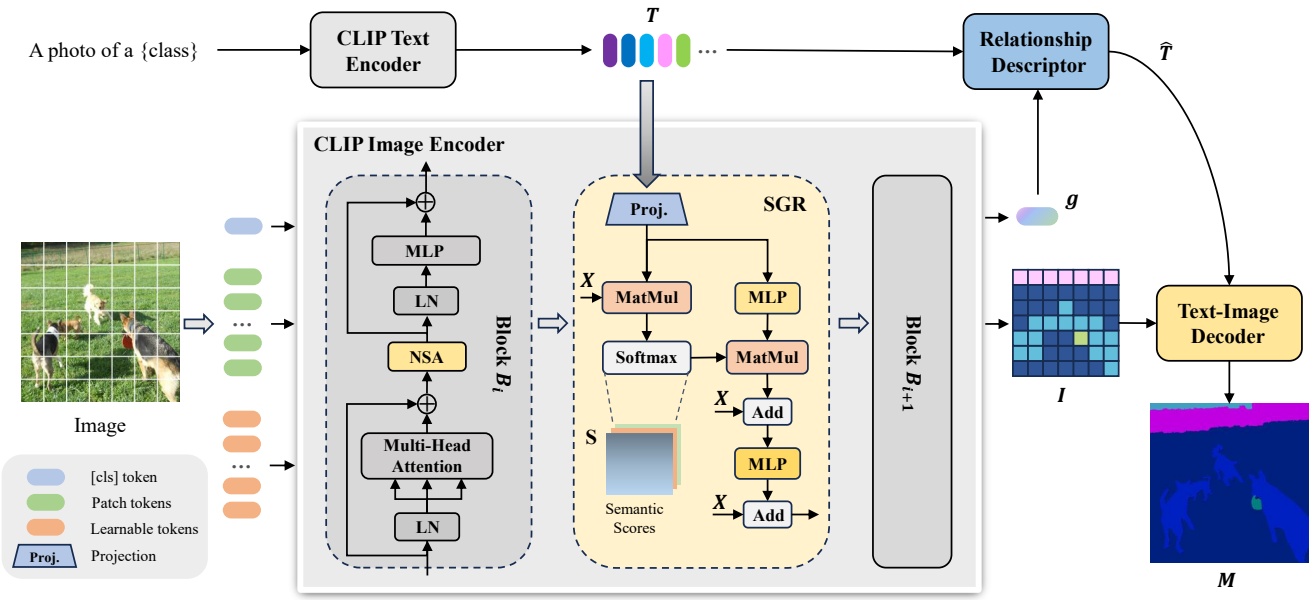

Figure 3. Overall of our proposed VR-CLIP. Our method is based on one-stage architecture that performs visual refinement within the CLIP visual encoder. The image is tokenized into patch embeddings and processed through NSA and SGR within and between the Transformer blocks of the CLIP visual encoder. Finally, the refined visual features I, together with the text representations $\hat{T}$ generated via Relationship Descriptor [47], are fed into a text–image matching decoder to produce the final prediction mask.

gating class imbalance between seen and unseen categories. With the advent of VLM, recent ZS3 methods have shifted to leveraging pre-trained cross-modal knowledge. Two-stage paradigms [9, 15, 28, 40], first generate class-agnostic region proposals and then use CLIP for image-level classification of cropped regions. These methods incur high computational overhead due to dual image encoding. To address this inefficiency, one-stage methods have been developed. ZegCLIP [47] introduced a lightweight transformer decoder to match text embeddings with CLIP's patch features directly, combining prompt tuning, non-mutually exclusive loss, and relationship descriptors to retain zero-shot capability. SPT-SEG [41] integrated spectral information to enhance the CLIP's generalization ability. Cascade-CLIP [23] further improved multi-level feature utilization by splitting CLIP's visual encoder into cascaded stages, each with an independent text-image decoder, enhancing alignment between intermediate visual features and text. OT-Seg [18] leverages Optimal Transport–based multi-prompt alignment to refine cross-modal matching. RC-CLIP [44] bridges the gap between image-level and pixel-level semantics via regional clues. Moreover, LDVC [45] introduces local–dense visual calibration in the decoder stage to enhance feature consistency, and TagCLIP [22] embeds a learnable trusty token into text prompts to strengthen discrimination between seen and unseen classes. [3, 11] introduces additional image encoders to enhance visual features.

Although the above methods successfully extend CLIP's image-level recognition capability to pixel-level segmentation, most of them rely on post-hoc processing after visual encoding. In contrast, our approach explores refining visual features during the encoding stage, to the best of our knowledge, is the first to explicitly optimize visual representations for ZS3.

3. Method

In this section, we begin by formulating the problem setting and revisiting the one-stage text–image segmentation paradigm for ZS3. Then, we provide a detailed description of our two key components.

3.1. Problem Definition

The zero-shot semantic segmentation task [1, 47] aims to segment both the seen classes $\mathcal{C}^s$ and unseen classes $\mathcal{C}^u$ after only training on a dataset with pixel-annotations of seen part, note that $\mathcal{C}^s \cap \mathcal{C}^u = \varnothing$. ZS3 is typically studied under two settings. In the inductive setting, only labeled data of seen classes is available during training, while the names and images of unseen classes remain completely unknown. This inevitably causes the model to be biased toward seen categories during inference, making it difficult to recognize unseen ones. In the transductive setting, the names of unseen classes are known during training, but their corresponding pixel-level annotations are not provided. Training images may contain regions belonging to unseen classes, yet these regions lack ground-truth segmentation masks.

3.2. Overview

Our work is based on the one-stage architecture [41, 47]. As shown in Fig. 3, given a set of C classes, their text embeddings are extracted as $T = [t_1, t_2, \ldots, t_c]^T \in \mathbb{R}^{C\times d}$. The image is tokenized into patch tokens $I \in \mathbb{R}^{N\times d}$, which are combined with a [cls] token $g \in \mathbb{R}^{1\times d}$ and several learnable tokens [41] before being fed into the CLIP visual encoder, where N denotes the number of patch tokens, and d denotes the embedding dimension. Within the encoder, our two proposed components perform visual refinements, and the final visual representation I is obtained from the output of the last layer. Notably, the CLIP visual encoder remains completely frozen during training, while only the parameters of our NSA and SGR are learnable. To mitigate overfitting, we adopt the Relationship Descriptor proposed in ZegCLIP [47] to optimize the text embeddings T:

$$\hat{T} = [\, T \odot g \,,\, T \,] \in \mathbb{R}^{C\times 2d}, \tag{1}$$

where $\odot$ denotes element-wise multiplication and $[\,,\,]$ indicates concatenation. The segmentation masks $M \in \mathbb{R}^{C\times N}$ are then obtained by computing the similarity between $\hat{T}$ and I within a text–image decoder:

$$M = \text{Softmax}\big(D(\phi_q(\hat{T}),\, \phi_k(I))\big), \tag{2}$$

where $D(\cdot)$ represents the text–image decoder, and ϕ_q, ϕ_k are linear projections that align the feature dimensions of $\hat{T}$ and I.

3.3. Neighborhood Spatial Awareness

In the original CLIP vision transformer, each residual attention block consists of a global attention layer and a feed-forward MLP, both followed by residual connections. While this formulation enables effective long-range dependency modeling, it tends to overlook fine-grained spatial relationships between adjacent patches, leading to weakened local consistency around object boundaries and textured regions. To address this issue, we insert a lightweight **N**eighborhood **S**patial **A**wareness module (Fig. 4) directly after the attention layer of the transformer block. NSA explicitly enhances local spatial coherence by integrating depthwise convolutional aggregation within the token space before the MLP stage, thus enriching neighborhood information for subsequent layers. The modified forward process is defined as:

$$\begin{aligned} X_a &= X + \text{Attention}(\text{LN}_1(X)), \\ X_s &= \text{NSA}(X_a), \\ X_{out} &= X_s + \text{MLP}(\text{LN}_2(X_s)), \end{aligned} \tag{3}$$

where $X \in \mathbb{R}^{N\times d}$ denotes the input patch tokens, LN_1, LN_2 denotes Layer Normalization.

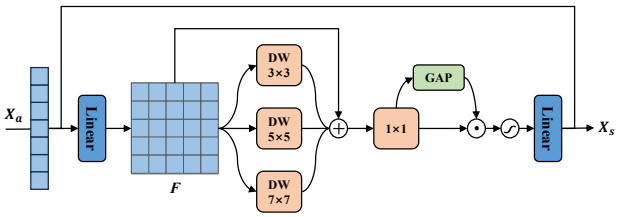

Figure 4. Detailed Description of the Neighborhood Spatial Awareness Module.

Specifically, we first project the patch tokens X_a into a compact latent space:

$$P = X_a W_1 + b_1, \quad P \in \mathbb{R}^{N\times D}, \tag{4}$$

where D is a reduced hidden dimension (e.g., 64), avoiding computational redundancy caused by excessive channels. We then reshape the sequence into a 2D feature map to restore spatial topology, enabling us to apply convolutions that aggregate spatial neighborhood information:

$$\begin{aligned} F &= \text{Reshape}(P) \in \mathbb{R}^{D\times H\times W}, \\ H \times W &= N. \end{aligned} \tag{5}$$

The neighborhood aggregation can be viewed as a multi-scale filtering process that enhances spatial smoothness:

$$F' = \psi\left(F + \sum_{k\in\{3,5,7\}} \alpha_k \cdot (F * \mathcal{K}_k)\right), \tag{6}$$

where $\mathcal{K}_k$ represents a depthwise convolutional kernel of size k, $*$ denotes depthwise convolution, α_k are learnable scale coefficients, and $\psi(\cdot)$ is a channel-mixing function implemented with a 1×1 convolution. This design ensuring channel-wise independent filtering and minimal parameter increase, reinforcing short-range spatial coherence within the feature map.

Then, feature weighting is performed on different channels through global average pooling and dot product operations. The enhanced features are then flattened, passed through nonlinear activation, and projected back to the original embedding dimension d:

$$P' = W_2\, \text{GELU}(\text{Flatten}(F')) + b_2, \tag{7}$$

after which a residual fusion yields:

$$X_s = X_a + P'. \tag{8}$$

To maintain efficiency, NSA is applied only to early transformer layers, where low-level spatial cues such as edges and textures are most informative.

3.4. Semantic-Guided Visual Refinement

Inspired by DenseCLIP [30], this approach enhances CLIP's dense prediction capability by incorporating pixel-level text score maps into the decoder. We further observe that CLIP's visual–language alignment is not confined to the image level but can also manifest at the pixel level. This observation motivates us to "see" images from a linguistic perspective, where enriched semantic cues are leveraged to refine visual representations more effectively.

Specifically, our approach performs layer-wise refinement of intermediate feature maps through **S**emantic-**G**uided Visual **R**efinement (Fig. 3) applied after each Transformer block of the frozen backbone.

Given an input image, the patch embedding layer outputs initial features $X \in \mathbb{R}^{N\times d}$. For the i-th frozen Transformer block B_i, its output is denoted as X_i. SGR then computes a semantic correction ΔX_i which is added to X_i to obtain the input of the next block:

$$X_{i+1} = B_{i+1}(X_i + \Delta X_i), \quad i = 1, \ldots, n-1, \tag{9}$$

and the final representation I is computed as:

$$I = X_n + \Delta X_n. \tag{10}$$

For each layer i, we introduce m semantic tokens $T_s \in \mathbb{R}^{m\times d}$ to interact with visual features. These tokens are obtained by linearly projecting the class embeddings T from a pre-trained CLIP text encoder:

$$T_s = \mathrm{LinearProj}(T). \tag{11}$$

Since the visual and text feature dimensions are not identical, this projection maps the text embeddings into the visual feature space, enabling semantic information from classes is incorporated into the initial token representations.

We adopted an approach similar to [30] to capture the association between semantic and visual information. Given the block output X, SGR computes patch–token similarities to capture semantic correspondence:

$$S = \mathrm{Softmax}\left(\frac{XT_s^\top}{\sqrt{d}}\right), \quad S \in \mathbb{R}^{N\times m}, \tag{12}$$

where Softmax is applied along the token dimension for each patch.

We then aggregate token information and produce a residual via two small MLPs:

$$\tilde{t} = \mathrm{MLP}_{\text{token}}(T_s) \in \mathbb{R}^{m\times d}, \tag{13}$$

$$\Delta X = \alpha \cdot \mathrm{MLP}_{\delta}\big(X + S\,\tilde{t}\big) \in \mathbb{R}^{N\times d}, \tag{14}$$

where α is a learnable scalar (initialized to a small value, e.g. 0.01) for stable training. $\mathrm{MLP}_{\text{token}}$ transforms the semantic tokens into a refined semantic basis that better aligns with the visual feature space, while MLP_{δ} adapts and reshapes the fused visual-semantic signal to produce an effective residual refinement.

In practice, to ensure effectiveness for unseen classes, $\mathrm{MLP}_{\text{token}}$ and MLP_{δ} are shared across layers. This design allows the MLPs to perform consistent linear transformations across different feature spaces within the backbone, maintaining a unified mapping of semantic information and residual refinement, which further enhances the model's generalization to novel categories. By applying SGR after each Transformer block, the model injects semantically informed, layer-specific refinements while preserving the frozen backbone's pre-trained visual–linguistic alignment. When combined with the spatial coherence enforced by NSA, these complementary refinements yield significantly more continuous and discriminative visual features, as illustrated in Fig. 2.

3.5. Loss Function

We employ a combination of the focal loss [25] and the structural similarity (SSIM) loss [33, 35]. The focal loss mitigates the class imbalance problem by focusing on hard examples, while the SSIM loss preserves structural consistency in the predicted segmentation maps. The total loss $\mathcal{L}$ is formulated as a linear combination of the two components, with coefficients α and β controlling their relative importance:

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{\text{focal}} + \beta \cdot \mathcal{L}_{\text{ssim}}. \tag{15}$$

Here, α and β are hyperparameters that balance the contributions of the two losses in the overall optimization objective.

4. Experiment

4.1. Datasets and Evaluation Metrics

To evaluate the effectiveness of our proposed method, we perform extensive experiments on three widely used benchmark datasets, including PASCAL-VOC 2012 [10], COCO-Stuff 164K [2], and PASCAL-Context [26]. The split of seen and unseen categories follows the common settings of the previous works [47]. The details of the datasets are described as follows:

PASCAL VOC 2012 consists of 10,582 training images and 1,449 validation images. Following common practice, the "background" category is excluded, using 15 classes as seen and 5 classes as unseen.

COCO-Stuff 164K is a large-scale benchmark with 118,287 training images and 5,000 test images, covering 171 categories. The dataset is divided into 156 seen classes and 15 unseen classes.

PASCAL Context contains 60 categories, with 4,996 images for training and 5,104 for testing. Among them, 50

Table 1. Comparison with the state-of-the-art ZS3 methods in ***inductive*** setting and ***transductive*** setting. R denotes ResNet. * denotes the results obtained from our retraining of the method. ST represents re-train the model with the generating pseudo-labels on unseen classes. **Bold** and underlined numbers indicate the best and second-best results, respectively.

Methods	Backbone	PASCAL VOC 2012			COCO-Stuff 164K		
		$\text{mIoU}^S\uparrow$	$\text{mIoU}^U\uparrow$	hIoU↑	$\text{mIoU}^S\uparrow$	$\text{mIoU}^U\uparrow$	hIoU↑
Inductive							
ZS3Net [1]	R101	77.3	17.7	28.7	34.7	9.5	15.0
CaGNet [13]	R101	78.4	26.6	39.7	33.5	12.2	18.2
ZegFormer [9]	R101&CLIP-B	86.4	63.6	73.3	36.6	33.2	34.8
Zsseg [39]	R101&CLIP-B	83.5	72.5	77.5	39.3	36.3	37.8
DeOP [15]	R101&CLIP-B	88.2	74.6	80.8	38.0	38.4	38.2
ZegCLIP* [47]	CLIP-B	90.5	78.3	84.0	40.2	41.4	40.8
LDVC [45]	CLIP-B	92.5	82.3	87.2	**43.2**	**45.0**	**44.1**
CLIP-RC [44]	CLIP-B	92.8	84.4	88.4	40.9	41.6	41.2
SPT-SEG* [41]	CLIP-B	93.1	85.4	89.1	39.7	40.3	40.0
TagCLIP [22]	CLIP-B	93.5	85.2	89.2	40.7	43.1	41.9
Cascade-CLIP* [23]	CLIP-B	92.8	83.0	87.6	40.8	42.6	41.7
VR-CLIP (Ours)	CLIP-B	**94.2**	**87.8**	**90.9**	42.2	42.8	42.5
Transductive							
ZS5Net [1]	R101	78.0	21.2	33.3	34.9	10.6	16.2
CaGNet+ST [13]	R101	78.6	30.3	43.7	35.6	13.4	19.5
Zsseg+ST [39]	R101&CLIP-B	79.2	78.1	79.3	39.6	43.6	41.5
FreeSeg [28]	R101&CLIP-B	91.9	78.6	84.7	42.4	42.2	42.3
MaskCLIP+ [46]	CLIP-B	88.8	86.1	87.4	38.1	54.7	45.0
ZegCLIP+ST* [47]	CLIP-B	92.5	89.9	91.2	40.5	59.6	47.8
LDVC+ST [45]	CLIP-B	-	-	-	41.1	60.3	48.8
CLIP-RC+ST [44]	CLIP-B	93.9	92.2	93.0	42.0	60.8	49.7
SPT-SEG+ST* [41]	CLIP-B	93.6	92.2	92.9	41.7	59.6	49.1
TagCLIP+ST [22]	CLIP-B	94.3	92.7	93.5	40.8	60.0	48.6
Cascade-CLIP+ST* [23]	CLIP-B	93.4	93.0	93.2	41.0	61.2	49.1
VR-CLIP+ST (Ours)	CLIP-B	**94.6**	**94.4**	**94.6**	**43.7**	**62.4**	**51.3**

Table 2. Comparison with the state-of-the-art ZS3 methods on PASCAL Context dataset.

Methods	PASCAL Context		
	$\text{mIoU}^S\uparrow$	$\text{mIoU}^U\uparrow$	hIoU↑
Inductive			
ZegCLIP [47]	53.8	45.5	49.3
SPT-SEG [41]	55.4	45.8	50.1
Cascade-CLIP [23]	55.9	47.2	51.2
VR-CLIP (Ours)	**57.5**	**48.6**	**52.7**
Transductive			
ZegCLIP+ST [47]	54.5	41.4	47.1
SPT-SEG+ST [41]	56.2	53.7	54.9
Cascade-CLIP+ST [23]	56.4	55.0	55.7
VR-CLIP+ST (Ours)	**58.8**	**56.7**	**57.7**

classes (including "background") are treated as seen, while the remaining 10 classes serve as unseen in the test set.

Following the previous works [23, 41, 47]. We evaluate performance with mean intersection over union (mIoU) for both seen and unseen categories, denoted as mIoU^S and mIoU^U. Additionally, we calculate the harmonic mean IoU (hIoU) between the seen and unseen classes, which is formulated as:

$$\text{hIoU} = \frac{2 * (\text{mIoU}^S * \text{mIoU}^U)}{\text{mIoU}^S + \text{mIoU}^U}. \tag{16}$$

4.2. Implementation Details

We implement the proposed method on the open-source toolbox MMSegmentation and conduct all experiments using a machine with two A800 GPUs. For the image encoder of CLIP, we adopt ViT-B/16, which consists of 12 Transformer layers. The batch size on each GPU is set to 8, and the input image resolution is 512 × 512. For the inductive setting, we use 20K training iterations on PASCAL VOC 2012, 40K on PASCAL-Context, and 80K on COCO-Stuff 164K. In the transductive setting, the model is initially trained on the seen classes during the first half of the iterations, and self-training with pseudo labels is applied during the remaining iterations. The optimizer is AdamW with the default training schedule in the MMSeg toolbox.

(a) Image (b) Ground truth (c) ZegCLIP (d) SPT-SEG (e) Ours

Figure 5. Qualitative results on COCO-Stuff164K dataset. Seen and unseen classes are distinguished by black and red tags, respectively.

4.3. Comparison with State-of-the-Art Methods

To comprehensively validate the effectiveness of our proposed VR-CLIP, we compare it with a wide range of zero-shot semantic segmentation methods under both inductive and transductive settings. Across three challenging benchmarks, our model achieves excellent performance on both seen and unseen classes.

Comparisons in the inductive setting. As summarized in Tab. 1. On PASCAL VOC 2012, VR-CLIP achieves an unseen mIoU of 87.8% and a hIoU of 90.9%, outperforming SPT-SEG [41] and Cascade-CLIP [23] by 1.8% and 3.3%, respectively. Similar trends are observed on the COCO-Stuff 164K datasets, where our model achieves a hIoU of 42.5%, outperforming SPT-SEG [41] by 2%. The same effectiveness is observed on PASCAL Context dataset, as shown in Tab. 2. These consistent gains highlight that refining CLIP's visual features from both semantic and spatial perspectives not only enhances its feature representation at the pixel level but also preserves its strong generalization ability.

Comparisons in the transductive setting. We further evaluate our model under the transductive zero-shot semantic segmentation setting. As reported in Tab. 1, our model consistently outperforms both inductive counterparts and other transductive baselines. Specifically, on PASCAL VOC 2012, our model with self-training achieves a hIoU of 94.6%, surpassing the previous best model Tag-CLIP+ST [22] by 1.1%. On PASCAL Context, our model also achieves the highest overall performance with a hIoU of 57.7%. Similarly, on COCO-Stuff 164K, our method yields a hIoU of 51.3%, marking the best performance among all compared methods and demonstrating effective utilization of unlabeled data for semantic refinement.

Qualitative results. Fig. 5 shows segmentation results from our proposed VR-CLIP method and previous approaches. VR-CLIP yields more complete and semantically consistent segmentations on unseen classes while accurately distinguishing visually similar classes. For example, 'clouds' over the 'sea' (Fig. 5(1)), shadows on the 'road' (Fig. 5(2)), especially the classes in low light at dusk (Fig. 5(3)). These results provide strong evidence of the superiority and robustness of our proposed framework.

4.4. Ablation Study

In this section, we perform ablation studies on the PASCAL VOC 2012 dataset under the inductive setting. We first examine the overall effectiveness of the two key components, and then conduct detailed analyses for each component individually.

Component analysis. Tab. 3 reports the effect of the two core components in VR-CLIP. Removing both SGR and NSA leads to the lowest performance. Adding NSA alone improves boundary consistency, yielding 89.8%

Table 3. Ablation on components of our method. SGR denotes Semantic-Guided Visual Refinement, NSA denotes Neighborhood Spatial Awareness.

SGR	NSA	mIoU^S↑	mIoU^U↑	hIoU↑
×	×	93.1	85.4	88.7
×	✓	93.4	86.8	89.8
✓	×	93.8	87.1	90.3
✓	✓	**94.2**	**87.8**	**90.9**

Table 4. Ablation on applying NSA to different layers of the CLIP visual encoder.

Insert Layer	mIoU^S↑	mIoU^U↑	hIoU↑
w/o NSA	93.1	85.4	88.7
Layer 1-12	93.2	36.9	52.9
Layer 1-6	93.2	83.3	88.0
Layer 1-2	**93.4**	**86.8**	**89.8**
Layer 11-12	93.1	84.3	88.5

hIoU, which is consistent with its design motivation of promoting local spatial continuity. SGR alone boosts hIoU to 90.3%, indicating that semantic-guided refinement serves as the dominant contributor by improving visual-linguistic correspondence. Combining both modules achieves the best result, demonstrating their complementary roles in improving semantic understanding and spatial precision.

Neighborhood Spatial Awareness (NSA). Applying NSA to all or most layers severely degrades performance on unseen classes, confirming that overly altering CLIP's pretrained representations undermines its original generalization capabilities. In contrast, inserting NSA only into early layers (Layer 1-2) achieves the best result. At this stage, features mainly encode low-level patterns such as edges, textures, and color transitions, which naturally benefit from localized convolution-based aggregation. Adding NSA to mid or late layers yields marginal improvement, as these layers focus on global semantics where convolution offers limited benefits.

Semantic-Guided Visual Refinement (SGR). Tab. 5 investigates the impact of the number of semantic tokens in SGR. Performance grows steadily as the token count increases, reaching the best result at 100 tokens. We argue that a moderate number of tokens provides sufficient semantic granularity to represent fine-grained concepts such as part-level shapes and contextual co-occurrence priors. Interestingly, we observe that too many semantic tokens slightly harms performance. This is likely due to redundant token interactions generating noisy similarity maps, which may affect the stability of residual refinement. These highlight that SGR not only injects high-level semantic information but also implicitly regularizes semantic sparsity during refinement.

Table 5. Ablation on the number of semantic tokens in SGR.

Number of Tokens	mIoU^S↑	mIoU^U↑	hIoU↑
20	93.6	85.8	89.5
40	**93.9**	86.2	89.9
100	93.8	**87.1**	**90.3**
160	93.8	84.8	89.1

Table 6. Effect of cross-Layer independent/shared parameter with MLPs in SGR.

Settings	mIoU^S↑	mIoU^U↑	hIoU↑
Independent	93.7	82.1	87.5
Shared	**93.8**	**87.1**	**90.3**

Effect of cross-layer parameter sharing. In SGR, two MLPs are used to integrate semantic cues into visual features: MLP$_{token}$ for semantic projection, and MLP$_{\delta}$ for residual refinement. We examine whether to share these parameters across layers or maintain independent MLPs. Tab. 6 shows that the shared configuration performs better, especially on unseen classes. Independent parameters cause each layer to learn separate semantic mappings, increasing the risk of overfitting to seen categories and breaking semantic consistency along depth. Sharing parameters not only stabilizes optimization but also promotes a unified refinement process across all layers. This design aligns with our philosophy that SGR should serve as a "global semantic controller" applied throughout the visual hierarchy. Qualitatively, we find that shared SGR produces more coherent activations around visually similar regions, supporting the notion that consistent semantic refinement is crucial for effective zero-shot generalization.

5. Conclusion

In this paper, we present VR-CLIP, a unified framework that refines CLIP's visual representations from semantic and spatial perspectives for zero-shot semantic segmentation. Instead of strengthening decoders or applying post-hoc adaptations, VR-CLIP performs in-encoder refinement, enabling fine-grained and semantically coherent pixel representations while preserving the backbone's visual–linguistic alignment. Extensive experiments show that VR-CLIP achieves strong performance in both inductive and transductive settings, producing more complete and consistent segmentations of unseen categories. By improving generalization without additional supervision, VR-CLIP takes a step toward open-world visual understanding with promising potential in applications such as autonomous driving and embodied intelligence.

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